ELEMENT I

Priorities

1) Programming- We had to test this by putting the battery at an angle and hooking it up with the gear.We tested this by pressing the push button.The battery was too strong and made the gear become loose and fly off.
2) Fit a 10x20 tray-We had to measure our box to make sure it fits.We had to measure the hole we cut out so we could easily slide the tray in.
3) Blinds that move with desired amount of sunlight-We couldn't find the appropriate volt battery.It was either too fast or too slow.

We tried to test it by connecting the battery and letting it open and close the blinds. This test was a high priority for us because this is all our project does. Without the blinds being able to open and close we just have a box with blinds on top. We cut the stick (what you twist to manually open the blinds) on the blinds down; so, we could connect a battery to it so when we push the button they will open and close. We tried to hook it up to a 6 volt battery and it was too weak and then we tried to put it on a 12 volt battery and it was too strong and too fast. We tried to use gears to connect the stick and the battery. After the first few attempts worked the last time before our design-a-thon the project would not work.We had to stay after and try to come up with solutions for our problem.We had to prop the battery up using a spare piece of wood and cut a straw to put on the stick so it would stay steady while turning the gears. The more and more we tried it would only work sometimes. Then we tried it the next day and the battery was spinning the gear so fast it made the gear fly completely off of the project.

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Independent Variable (Blinds)
The independent variable is the blinds; it stays fixed in the same position. It is dependent on the gears; which rotate clockwise and counterclockwise which in turn opens and closes the blinds.

Dependent Variable(Angle of Rod)
We were able to change the angle of the rod to change the speed and how fast it would open and close.

We changed the angle of the rod of blinds that open and close 5 times.
1. At first we positioned the rod at a 90 degree angle pointing downwards. We tested it out and it had no effect on the blinds.
2. This time we positioned the rod; so it was at a 70 degree angle. Upon testing, the blinds took too long to open and close
3. The third time we put the rod at a 60 degree angle; it opened quickly but it did not close at all.
4. The fourth time we angled the rod at a 50 degree angle; it closed perfectly, but it did not open efficiently
5. The last time we positioned the rod at a 45-40 degree angle; it opened and closed a bit more slowly but overall it was the most efficient out of the last 4 attempts.

Summary of Tests: From our 5 attempts we can conclude that the main problem was the angle of the rod. There was a difference in the effectiveness of the blinds opening and closing depending on the angle of the rod.

ELEMENT I

For this project we have not yet found the correct battery to use.We didn’t get to fix our project.When we do try and fix it we will try and use a 9 volt battery.Trying to fix it using the gears was a good idea.We just need a slower battery.

Unfortunately for our project all the times we tried to use the battery to spin the blinds it worked except for the last time before our design-athon.When we had tried it before it worked fine.It was a little fast but we could deal with it.When we tried it the last time it wouldn't open and close. So we attached gears; but, slowly after trying it a few times the gear came off.